

Decent Coaching Classes
X Semi
Mind Your Math - VI — Answer Key

Q.1 (A)

- (1) (B) (3, 2) $[3 + 2 = 5 \text{ and } 3 - 2 = 1]$
(2) (B) 18 $[6 \times 5 - 4 \times 3 = 30 - 12 = 18]$
(3) (C) $x^2 - 5x + 6 = 0$
(4) (B) 3

Q.1 (B)

- (1) $2y = 10 - y^2 \Rightarrow y^2 + 2y - 10 = 0$
(2) $a = 1, b = 7, c = -1$. $b^2 - 4ac = 49 - 4(1)(-1) = 49 + 4 = 53$
(3) $t_1 = 10, t_2 = 15, t_3 = 20, t_4 = 25$. A.P. : 10, 15, 20, 25, ...
(4) First term $a = 5$; common difference $d = 1 - 5 = -4$

Q.2 (A) (any two)

- (1) Adding (I) and (II) : $6x = 24 \therefore x = 4$
Putting $x = 4$ in (I) : $3 \times 4 + 2y = 16 \therefore 2y = 4 \therefore y = 2$
 $\therefore$ Solution is $(x, y) = (4, 2)$
(2) $x^2 - 3x - 4x + 12 = 0$
 $x(x - 3) - 4(x - 3) = 0$
 $(x - 3)(x - 4) = 0 \therefore x = 3 \text{ or } x = 4$
(3) $t_1 = 3, t_2 = 8, t_3 = 13, t_4 = 18$
 $t_2 - t_1 = 5, t_3 - t_2 = 5, t_4 - t_3 = 5$
The difference is constant, so it is an A.P. with $d = 5$. Next two terms are 23 and 28.

Q.2 (B) (any two)

- (1) For $x = 1$: $1^2 + 4(1) - 5 = 1 + 4 - 5 = 0$. $\therefore x = 1$ is a root.
For $x = -1$: $(-1)^2 + 4(-1) - 5 = 1 - 4 - 5 = -8 \neq 0$. $\therefore x = -1$ is not a root.
(2) $|-1 \ 7 ; 2 \ 4| = (-1)(4) - (7)(2) = -4 - 14 = -18$
(3) Sum of roots = $3 + (-10) = -7$; Product of roots = $3 \times (-10) = -30$
Equation : $x^2 - (\text{sum})x + (\text{product}) = 0 \Rightarrow x^2 + 7x - 30 = 0$

Q.3 (A) (any one)

- (1) $a = 1, b = 6, c = 5$
 $b^2 - 4ac = 36 - 20 = 16$
 $x = (-6 \pm \sqrt{16}) / 2 = (-6 \pm 4) / 2 \therefore x = -1 \text{ or } x = -5$
(2) $D = (1)(-3) - (2)(2) = -3 - 4 = -7$
 $D_x = (-1)(-3) - (2)(12) = 3 - 24 = -21$
 $D_y = (1)(12) - (-1)(2) = 12 + 2 = 14$
 $x = D_x / D = -21 / -7 = 3$; $y = D_y / D = 14 / -7 = -2$
 $\therefore$ Solution is $(x, y) = (3, -2)$

Q.3 (B) (any two)

- (1) $3a + 5b = 26 \dots (I)$; $a + 5b = 22 \dots (II)$
Subtracting (II) from (I) : $2a = 4 \therefore a = 2$

Putting $a = 2$ in (II) : $2 + 5b = 22 \therefore 5b = 20 \therefore b = 4$

$\therefore$ Solution is $(a, b) = (2, 4)$

(2) $x^2 - 15x + 54 = 0 \Rightarrow x^2 - 6x - 9x + 54 = 0$

$$x(x - 6) - 9(x - 6) = 0 \Rightarrow (x - 6)(x - 9) = 0$$

$$\therefore x = 6 \text{ or } x = 9$$

(3) Here $a = 1, b = -4, c = 4$

$$b^2 - 4ac = 16 - 16 = 0$$

Since the discriminant is 0, the roots are real and equal.

Q.4 (any one)

(1) $x + y = 6$: $x = 0 \Rightarrow y = 6$; $x = 3 \Rightarrow y = 3$; $x = 6 \Rightarrow y = 0 \rightarrow$ points $(0, 6), (3, 3), (6, 0)$

$x - y = 4$: $x = 0 \Rightarrow y = -4$; $x = 4 \Rightarrow y = 0$; $x = 5 \Rightarrow y = 1 \rightarrow$ points $(0, -4), (4, 0), (5, 1)$

Plot the points and draw both lines on graph paper; they intersect at $(5, 1)$.

$$\therefore \text{Solution is } x = 5, y = 1.$$

(2) Let the smaller number be x and the greater number be y .

$$y - x = 3 \dots \text{(I)} \text{ and } 2x + 3y = 19 \dots \text{(II)}$$

From (I) : $y = x + 3$. Putting in (II) : $2x + 3(x + 3) = 19 \Rightarrow 5x + 9 = 19 \Rightarrow x = 2$

$$\therefore y = 2 + 3 = 5. \text{ The numbers are 2 and 5.}$$

(3) Let Pragati's present age be x years.

$$(x - 2)(x + 3) = 84 \Rightarrow x^2 + x - 6 = 84 \Rightarrow x^2 + x - 90 = 0$$

$$x^2 + 10x - 9x - 90 = 0 \Rightarrow x(x + 10) - 9(x + 10) = 0 \Rightarrow (x + 10)(x - 9) = 0$$

$$\therefore x = -10 \text{ or } x = 9. \text{ Age cannot be negative, so } x = 9.$$

$$\therefore \text{Pragati's present age is 9 years.}$$